BRIDGES IN MATHEMATICS • GRADE 1 • UNIT 2

Developing Strategies with Dominoes & Doubles

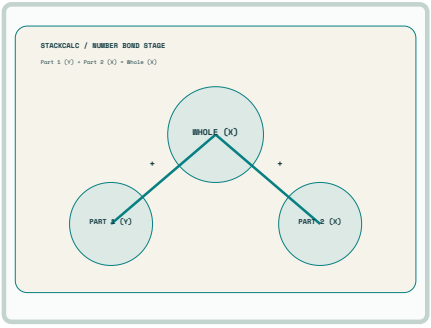
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Bond Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place double dominoes (4|4, 5|5) into the tray. Add 1 loose peg for near doubles.

Sketch Model on Paper:
Draw double domino dots with +1 callout circle.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	6	X: 6	Key 6 into X
2.	ENTER	Y: 6 | X: 6	Stack lift copies 6
3.	+	X: 12	Double 6: 6 + 6 = 12
4.	1	Y: 12 | X: 1	Key +1 into X

EXPECTED LCD X: 13 (Double 6 + 1)

1 **Use double 3 to solve 3 + 4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Use double 6 to solve 6 + 7.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Use double 8 to solve 8 + 9.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **What is double 9?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 2

Developing Strategies with Dominoes & Doubles • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Math Domino Studio (12-Frame Tray)

CAD Blueprint: number_bond_stage

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/math_dominoes_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Stage with White Dominoes (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Use double 3 to solve 3 + 4.** Standard: CCSS.MATH.1.OA.C.6	7	[3] [ENTER] [+] [1] [+]
2	**Use double 6 to solve 6 + 7.** Standard: CCSS.MATH.1.OA.C.6	13	[6] [ENTER] [+] [1] [+]
3	**Use double 8 to solve 8 + 9.** Standard: CCSS.MATH.1.OA.C.6	17	[8] [ENTER] [+] [1] [+]
4	**What is double 9?** Standard: CCSS.MATH.1.OA.C.6	18	[9] [ENTER] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students relate 4+5 back to known double 4+4=8.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.